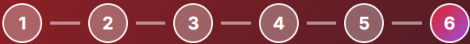


Learning Week: Day 1 → Day 6

A complete guide to understanding chains, agents, memory, orchestration, and building a real retail data pipeline



Day 1: Chains

Day 2: Templates

Day 3: Memory & Tools

Day 4: LangGraph

Day 5: Orchestrator

Day 6: The Project

Day 1: What is LangChain?

Chains vs. Raw API Calls

What is a Chain?

A **chain** is a fixed, step-by-step pipeline where data flows in a straight line from start to finish. Instead of calling an LLM API directly every time, LangChain gives you reusable building blocks.

WITHOUT LANGCHAIN (RAW API):

```
prompt = "Explain what is a chain"
response = llm.call(prompt)
# You write the full prompt every time
# Raw text back - hard to parse
```

WITH LANGCHAIN (CHAIN):

```
chain = prompt_template | llm | output_parser
result = chain.invoke({"topic": "chains"})
# Reusable, structured, reliable
```

Key Concepts

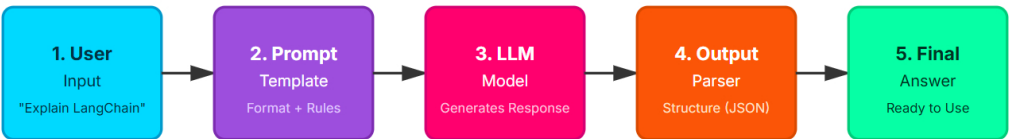
LLM Wrapper: A clean interface to any LLM (GPT, Gemini, Groq)

Prompt Template: A reusable prompt with {variables} you fill in

Chain: Connected components (template → LLM → parser)

LCEL: LangChain Expression Language - uses | to connect pieces

The 5-Step Chain Pipeline



ASPECT	RAW API CALL	LANGCHAIN CHAIN
Reusability	Write prompt every time	Template reusable
Output Structure	Free-flowing text	Structured (JSON/Pydantic)
Switching LLMs	Rewrite entire code	Change one line
Speed	Fast, single step	Slightly slower (structured)
Flexibility	Fixed behavior	Fixed behavior (predictable)

Key Takeaway: Chains are predictable, reusable, and reliable. But they're rigid — they always follow the same path. If an LLM doesn't know something, a chain can't decide to search the web or try a different approach. That's where agents come in (we'll see this in Day 5).

Free Resources to Dive Deeper

LangChain Docs

Official documentation with API reference and guides

LangChain Academy

Free official courses on LangChain fundamentals

LangGraph Docs

Guide to building graphs and agents

Multi-Agent Tutorial

Real example of orchestrator agents in action

🌟 Learning Week: LangChain + LangGraph Fundamentals

DevSynt AI Automation Internship · Mentor: Afnan Shoukat

Quiz coming next — make sure you understand these concepts!